

PropFlo WhatsApp Automation System Design

1. Problem Understanding :

PropFlo manages multiple real estate projects under a single WhatsApp Business Account.

The goal is to design a system that:

- Handles **new and existing leads**
- Supports **multiple projects**
- Routes users to the **right sales agent**
- Enables smooth **handover to human agents**
- Follows **WhatsApp rules (24-hour window, templates)**

Core Problem Insight :

The main challenge is:

Mapping user intent to the correct project and agent while asking minimum questions and reducing drop-offs.

Current Problems :

- Slow response → users lose interest
- Manual handling → inconsistent experience
- Wrong routing → poor conversion
- No proper follow-up

2. Business Goal :

The primary goal of this system is to **increase lead conversion (from enquiry to site visit/booking)**.

This is achieved by:

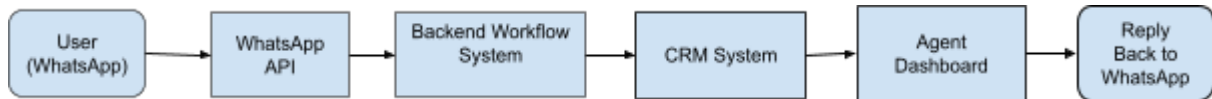
- Faster response time
- Better lead qualification
- Efficient routing to the right agent

3. Assumptions

- One builder with multiple projects
- Users come from **ads and direct messages**
- A CRM system already exists
- Agents reply using a dashboard
- WhatsApp rules must be followed

4. System Overview

Architecture Flow :



WhatsApp Integration Overview :

The system integrates with the **WhatsApp Business Platform** using its Cloud API.

- Incoming messages are received via **webhooks**
- Each message or button click triggers an **event**
- Backend processes the event and decides the next step
- The system maintains user **session/state** to track conversation progress
- Replies are sent using:
 - > Text message
 - > Reply buttons
 - > List messages
 - > Template messages (after 24 hours)

This is an **event-driven system** enabling real-time responses and scalability.

Execution Layer :

The workflow can be implemented using tools like **n8n**, enabling:

- Webhook handling
- Workflow-based logic
- CRM integration
- Automated responses

5. End-to-End WhatsApp Flow



Step-by-Step Flow :

Step 1: Entry Point

Ad Entry

- Project already known
- Bot confirms project

Organic Entry

- Ask:
“Which project or location are you looking for?”

Step 2: Identify Lead

- Check phone number in CRM
- Existing → fetch data
- New → create lead

Step 3: Project Selection

- From ad (auto-selected)
- Show project list
- Keyword-based match

User can switch anytime

Step 4: Ask Key Details

- BHK
- Budget
- Timeline
- Action (site visit / callback)

Use buttons (not typing)

Step 5: Agent Routing

- Skill-based (project-specific)
- Load balancing
- Round-robin fallback (If no special conditions apply, assign leads in a **fixed rotating order**)

Step 6: Human Handover

Triggered when:

- User asks for agent
- High intent
- Bot cannot help

Agent receives:

- User details
- Chat summary
- Conversation history

Step 7: 24-Hour Messaging Logic

- Within 24 hours → normal replies
- After 24 hours → only template messages

Example template:

“Hi, are you still interested in Project A? Let me know if you’d like to schedule a site visit.”

Step 8: Edge Cases

- Unclear input
- Duplicate lead
- Project change
- No agent available
- 24h window expired

Handling Edge cases:

Unclear input

→ Show retry prompt with buttons (“Did you mean...?”) + fallback to agent after 2 failed attempts

Duplicate lead

→ Merge with existing CRM record using phone number + update latest interaction

Project change

→ Update selected project in state + asking some of the key questions again when the user changes something important (like project).

No agent available

→ Queue the lead + notify next available agent + send user “We’ll connect shortly” message

24h window expired

→ Trigger approved template message for re-engagement

API + Event Logic:

- Each message triggers a **webhook**
- Button clicks update user state
- Backend decides next step dynamically

Ensures **real-time automation and scalability**

6. Conversation Design (UX)

Principles

- Ask fewer questions
- Use buttons/lists
- Avoid typing
- Keep flow simple

Ad Flow Example

1. Confirm project
2. Select BHK
3. Select budget
4. Select timeline

5. Book visit

Organic Flow Example

1. Ask project
2. Show list
3. Ask BHK
4. Ask budget
5. Route to agent

7. CRM Data Model

Entities :

Lead	Project	Conversation	Agent	Interaction Logs
Name, Phone	Name	Last Message Time	Project Handled	Messages
Project	Location	Current step	Availability	Button Clicks
Budget,BHK	Price Range	24h status	Load	Templates
Timeline				
Status				

AI-Ready Design :

The system stores structured data which enables:

- Lead tagging (Automatically label leads based on behaviour)
- Lead scoring (Give each lead a **score (priority level)**)
- Chat summaries (AI generates a short summary of the conversation)
- AI-powered improvements (System becomes smarter over time)

Data collected in Q1 will be used to **train AI models in later phases**

8. Multi-Project Scalability

This system won't break when we add more projects, users, and traffic.
The system can handle many real estate projects at the same time without redesign.

The system scales because:

- Projects are dynamic (You can add/remove projects easily)
- List-based selection
- Routing is project-aware (Which agent handles which project)
- Modular workflows (The system is built in small reusable pieces (modules))

Real-World Scalability

- Supports **100+ projects**
- Handles **high concurrent chats**
- Uses **event-driven system**

- Uses **queue-based routing**
- Supports **project-specific queues**

9. Product Roadmap

Q1 – Core System

- Rule-based chatbot
- CRM integration
- Lead identification
- Routing & handover
- WhatsApp compliance

Q2 – AI + Analytics

- AI tagging
- Lead scoring
- Funnel tracking
- Chat summaries

Q3 – AI Conversational Layer

- Smart chatbot
- Intent detection
- Personalization
- AI agent assistant

10. Success Metrics

KPIs

- First response time
- Lead qualification rate
- Drop-off rate
- Agent efficiency
- Site visit bookings

North Star:

“ Leads converted to site visits “

11. Final Summary

This system provides a scalable WhatsApp automation solution for PropFlo.

It:

- Handles new and existing leads
- Supports multiple projects
- Uses structured flows
- Ensures smooth handover
- Follows WhatsApp rules

Over time, it evolves into an **AI-powered system** for better personalization and automation.